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The Use of Peer Mediation and Educator Facilitation to Promote Turn Taking in Young Children With Autism Spectrum Disorder in Inclusive Childcare

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A multiple-probe design across participants was used to investigate the effect of a peer-mediated intervention on the turn-taking behavior of 3 children with autism spectrum disorder (ASD) attending inclusive childcare centers. An educator trained peers to support the child with a disability to take turns playing a game on an iPad. The educator was also available to provide coaching for the peer when needed. An intervention effect was demonstrated for each of the children, and 2 were able to demonstrate maintenance of the skill when the training and coaching procedures were not implemented by the educator. Although some elements of the treatment procedures were not consistently applied by the educators or typical peers, the intervention was sufficiently robust to enable the children with ASD to improve their turn-taking skills. **Key words:** *autism, childcare, coaching, peer mediation, single-case research, touch screen technology*

INCLUSION IN TYPICAL early childhood settings is considered to be the right of every child, including those with significant disabilities, and has also been identified as having benefits for both children with disabilities and their typically developing peers (Odom, Buysse, & Soukakou, 2011; Stahmer & Carter, 2005). Although no one would deny the right of children and families to access typical early childhood services such as preschools and childcare, most would acknowledge that successful inclusion should involve active intervention to promote engagement, social participation and child development (Odom et al., 2011). A range of strategies, including peer-mediated interventions to support social-communication behavior, has been identified as potentially contributing to successful early childhood inclusion (Goldstein, Lackey, & Schneider, 2014). Turn taking with peers is one example of social behavior that would be expected of a 4- to 5-year-old child in an early childhood setting.

TARGETING TURN-TAKING SKILLS IN EARLY INTERVENTION

Turn-taking skills rely on joint attention to an object, person, or event, which is important to the development of early language and social skills (Lawton & Kasari, 2012; Toth, Munson, Meltzoff, & Dawson, 2006). Across developmental and intellectual levels, children with autism spectrum disorder (ASD) have difficulty developing skills in joint attention (Schietecatte, Roeyers, & Warreyn,

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2012) compared with their typically developing peers and also compared with children with delayed development (Kasari, Gulsrud, Wong, Kwon, & Locke, 2010; Lawton & Kasari, 2012; Toth et al., 2006). For typically developing children, turn-taking skills will emerge without systematic intervention. However, for children with ASD, an instructional program specifically targeting these skills may be necessary. Having educators provide initial training and ongoing coaching to typically developing peers to support the turn-taking behavior of children with ASD (i.e., a peer-mediated intervention) is one way in which such a program can be provided.

THE EFFICACY OF PEER-MEDIATED INTERVENTIONS IN EARLY CHILDHOOD SETTINGS

Peer-mediated interventions generally involve teaching typical peers to initiate and respond to the social interactions of children with ASD, thereby increasing the quality of these interactions (Zagona & Mastergeorge, 2018). As with joint attention, peer interaction skills are generally poorer in young children with ASD even when compared with children with other types of disability (Kemp, Kishida, Carter, & Sweller, 2013); therefore, it is important that these skills are promoted in inclusive preschools and childcare (Katz & Girolametto, 2013).

Since the 1970s, the use of peer-mediated interventions to support the development of social skills in children with disabilities, including young children with a diagnosis of ASD, has had support in the research literature, particularly through single-case experimental research (Odom et al., 2003; Rogers, 2000). In a review of social skills interventions for preschool children with ASD, Goldstein et al. (2014) found that 22 of the 67 identified studies involved peers and all but one of these studies employed a single-case experimental design. These interventions included social skills training for peers alone, social skills training both for peers and for children with disabilities within a small group or class, and group games. Strong empirical evidence was found for the use of peer-mediated social skills interventions for preschool children with ASD.

A review of peer-mediated interventions for children with ASD between 2004 and 2014 (Zagona & Mastergeorge, 2018) found that of the 17 studies that met the criteria for inclusion in the review, only two included children younger than 5 years. Only one of these two studies met the minimum standard for quality of research design. Similarly, a review of peer-mediated social interactions in inclusive settings, 2008 to 2014, by Watkins et al. (2015) identified that only three of the 13 studies included preschool-aged children. The authors of both reviews recommended that there should be more research into the use of peer-mediated interventions targeting social communication skills with preschool-aged children with ASD.

Additional research is needed to determine the level of training required by teachers to implement peer-mediated interventions with young children with ASD, as much of the research thus far has been implemented using research assistants or highly qualified teachers, who receive ongoing consultation, coaching, and support throughout the implementation of the intervention (Katz & Girolametto, 2013). Although there are studies that have indicated that this approach can be used successfully to increase the skills of educators in special settings (e.g., Fox, Hemmeter, Snyder, Binder, & Clarke, 2011) and typical childcare centers (e.g., Cain, Rudd, & Saxon, 2007; Kishida & Kemp, 2010), access to intensive training and ongoing support may not be available for educators in typical settings. It would, therefore, be important to determine whether evidence-based strategies, such as peer-mediated interventions, can be implemented reliably by educations without the need for intensive support.

ENGAGING PRESCHOOLERS USING TOUCH SCREEN DEVICES

Touch screens are now commonly available in the home (Bedford, Saez de Urabain, Cheung, Karmiloff-Smith, & Smith, 2016) and

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are accessed by children from a very young age (Chmiliar, 2017; Lovato & Waxman, 2016). There is evidence to suggest that there are developmental benefits for young children in the use of such technologies and that these are intrinsically attractive and engaging to young children, including children with ASD (Bedford et al., 2016; Fletcher-Watson et al., 2016). Mobile technologies such as iPads, mobile phones, and iPods are being used more often with children with a broad range of learning needs (Draper Rodriguez, Strnadová, & Cummings, 2014), including children and adults with developmental disabilities (Stephenson & Limbrick, 2015).

Single-case experimental designs have been used to demonstrate the value of mobile technology in the development of skills in children with significant disabilities. Skills targeted have included daily living skills (Wu, Canella-Malone, Wheaton, & Tullis, 2016), imitation (Cardon, 2012), communication (Fletcher-Watson et al., 2016; King et al., 2014; Waddington et al., 2014), academic engagement (Hart & Whalon, 2012; Neely, Rispoli, Camargo, Davis, & Boles, 2013), literacy (Kagohara, Sigafoos, Achmadi, O'Reilly, & Lancioni, 2012), and numeracy (Weng & Bouck, 2014). Although recently there has been a greater focus in the literature on the use of touch screen devices with young children with and without disabilities, there are few experimental studies targeting the use of technology with preschool children with significant disabilities. Single-case experimental designs have been used to demonstrate positive effects for the use of iPads in instructional programs designed to teach imitation (Cardon, 2012) and communication skills (Ganz, Hong, & Goodwyn, 2013; King et al., 2014) and also to promote the engagement (Kemp, Stephenson, Cooper, & Hodge, 2016) of preschool children with ASD. It is clear from the research, therefore, that iPads are attractive to preschoolers and have the potential to engage a wide range of children, including children with ASD.

THE PURPOSE OF THIS STUDY

The current study involved peer mediation, facilitated by educators, to teach early turn-taking skills using an iPad game to three preschool children with a diagnosis of ASD. The researchers relied on a 2-hr training session and carefully scripted instructions, but no ongoing support, to assist educators to implement peer training and coaching. It was important to determine whether this lower level of support would be sufficient to allow educators to implement the intervention with an acceptable level of reliability, which in turn would assist the peers to support successful turn taking in the children with ASD.

The following research questions guided the implementation of the research, the first question being the primary one:

1. Will an educator-implemented intervention involving the initial training and subsequent coaching of typically developing peers to support the turn-taking skills of children with disabilities result in an increase in these children's successful turn taking?
2. Will turn taking continue at a high level in sessions when training and coaching are not provided?
3. With minimal training, can educators implement the training and coaching program developed by the researchers with an acceptable level of fidelity?
4. Can typical peers implement the intervention with training and support from educators?

METHODS

Participants and settings

Three children with a diagnosis of ASD, 11 typically developing peers, and two educators participated in the research. The childcare centers in which the children were enrolled were affiliated with the STaR Childcare Support Program, which supports young children

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with disabilities, their families, and educators in approximately 10 childcare centers in Sydney, Australia. Before enrolling their children, the parents of the children with disabilities attending the STaR-affiliated centers agreed to participate in studies related to improved interventions for their children. Pseudonyms are used for both child and adult participants.

Children with ASD

All three children attended privately operated childcare centers in northwestern Sydney, Australia. Both centers were supported by an early childhood special educator, who provided both a professional development component and a coaching component to assist educators to use activity-based assessment and parent input in order to plan, implement, and evaluate an individual program for each child with a disability or delay attending the centers.

The three preschoolers (all boys) had been identified by the educator in their rooms as having difficulties with turn taking. All three had priority goals relating to social interaction and turn taking as part of their individual education plans. Kenny and Noah attended the same childcare center for 3 days a week and were in the same preschool group. Adam attended a second center for 2 days and later 3 days a week. Adam and Noah had been assessed by multidisciplinary teams at state government assessment centers, and Kenny had been assessed by a pediatrician at a private clinic.

Kenny

At the time the intervention began, Kenny was 56 months of age and had attended the childcare center for 19 months. He had been diagnosed with a mild developmental delay, a severe language disorder, and ASD. As reported by the educator in his room, Kenny enjoyed playing with construction materials during indoor free play and with bikes during outdoor play but was unable to maintain focus for whole group activities. Kenny was using short sentences but frequently spoke using movie scripts and was echolalic. The educator reported that he did not always tolerate having other children around while engaged in his favored play activities and would sometimes become aggressive with peers who interrupted his play.

Adam

At the commencement of the intervention, Adam was 54 months of age and had attended the childcare center for 31 months. He had been diagnosed with severe global delay and ASD. During free play activities, Adam was reported by the educator in his room to enjoy playing with play dough and sand. He spoke using single words and was being taught to use key signs to accompany them. He was able to remain on task for only a short period of time, with one of his objectives being to remain in supported play for 3 min. A priority goal for Adam was to increase social interaction with his peers.

Noah

Noah had attended his childcare center for the least amount of time (12 months). He was 57 months of age at the commencement of the research. Noah had been diagnosed with a moderate global developmental delay, ASD, and severe anxiety. According to the educator in his room, Noah was able to engage in independent play for short periods. He frequently watched other children playing and was beginning to tolerate having children around him. Although he had many single words in his receptive vocabulary, he had little expressive language. Noah was reported to cry or scream when there were changes in routine and to engage in task avoidance. Noah would often order numbers and letters and appeared to derive comfort from this activity.

Typical peers

Peers were chosen because they were in the same room (i.e., were of similar age) as the child with ASD and attended on the same days. The aim was to recruit four peers for each target child to allow for peer absence or a preference on the part of a recruited peer

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not to join the activity. Parents of children who either played alongside or showed interest in the children with ASD were approached to ask for permission for their children to participate in the research. The peers who finally participated were those for whom informed written consent from parents was received. See Table 1 for information relating to the peers supporting each of the target children over the course of the research.

Educators

Two childcare educators (one male and one female), one from each center, provided written consent to participate in the research. Sue was the educator for two of the children, Kenny and Noah, and Tim was the educator for Adam. Each educator worked in the room in which the children they supported were enrolled. Sue was a qualified early childhood teacher with a bachelor's degree and had worked at the center attended by Kenny and Noah for approximately 4 years. Tim had a degree in visual arts and a childcare certificate from a Technical and Further Education (TAFE) college. Tim had been working at the center for approximately 8 years.

Materials and equipment

An iPad app was used to engage the children in the turn-taking exercise. The selection of the app Peekaboo HD (Version 1.6; Gotclues Inc., 2013) was based on its success in engaging a child with a severe disability and ASD in an earlier study (Kemp et al., 2016). The game involved touching a habitat or cover to expose an animal, listening to the sound the animal made, and touching the habitat again to hide the animal. There are three versions of this game: farm, jungle, and safari. These versions were rotated across sessions to maintain the interest of the children.

A video camera, operated by administrative assistants specifically employed for this purpose, was used to film all baseline, intervention, maintenance, and training sessions. The data were then downloaded to a computer so that the footage could be coded for successful

Table 1. Peers Supporting Turn Taking

Noah			Adam			Kenny		
Peer	Baseline Sessions	Intervention/ Maintenance Sessions	Peer	Baseline Sessions	Intervention/ Maintenance Sessions	Peer	Baseline Sessions	Intervention/ Maintenance Sessions
Lucie (F)	B1, B4	I1, I3, I5	Alan (M)	B1, B3	I1, I5, I10, I13, I17	Sarah (F)	B1	I3, I4, I10
Harry (M)	B2, B3, B5	I2, I4	Terry (M)	B2	I2, I6, I8, I14, Ma1	Amalie (F)	B2	I6, I9, I12, I15, Ma1, Ma6
			Chloe (F)	B5	I3, I11, I12, I15, I16, I18	Gerald (M)	B3	I5, I8, I14, I19, I20, Ma2
			Amy (F)	B4	I4, I7, I9, I19	Jacinta (F)	B4	I1, I11, I16, I17, Ma3, Ma5
						Hannah (F)	B5	I2, I7, I13, I18, Ma4, Ma7

Note. B = baseline; F = female; I = intervention; M = male; Ma = maintenance.

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turn taking and fidelity of implementation by educators and peers.

Research design

A single-case, multiple-probe design across participants was used to determine the efficacy of a child-mediated approach to promoting turn-taking skills in young children with ASD. Turn-taking data collected from baseline and intervention phases were graphed for each participant and analyzed by visual inspection (Kratochwill et al., 2013). Maintenance data, that is, data collected in sessions for which peers were not trained or coached, were collected once participants had reached the target of 80% success for turn-taking opportunities.

Dependent variable

The dependent variable was successful turn taking. A successful turn was recorded when the child with ASD (a) opened and closed the animal cover or allowed the cover to close on its own; (b) took a turn first or took a turn following the peer's turn; and (c) allowed the peer to take his or her turn (open and close). Because young children sometimes need to be reminded to take a turn when participating in a game, the decision was taken to score the child as successfully taking his turn if this were done spontaneously or following a verbal prompt from his peer. An unsuccessful turn was recorded when the child with ASD (a) took a turn that should have been taken by his peer or tried to take his peer's turn; (b) did not take a turn following his peer's turn; (c) required a physical prompt to take his turn; and (d) closed the cover when it was his peer's turn or allowed his peer to close the cover when it was his turn. If a child opened the animal cover when it was his or her turn but allowed the other child to close it, this was considered to be the turn of the child who opened the cover. If a child opened the animal cover when it was not his or her turn and the other child closed it, the child who closed the cover was expected to take the next turn.

Independent variable

Typical peers were instructed by their educator to use a prompt hierarchy to assist the child with ASD to take turns using an iPad app. In the first instance, the peer was to take his or her turn and then wait for 3 s (i.e., count to three slowly) before providing a verbal prompt (e.g., "Adam, it's your turn now"). The peer was then to wait another 3 s before providing physical and verbal prompts. For this step, the child was instructed to gently take his friend's index finger and help him to touch the animal cover. If the child with ASD tried to take his turn out of order, the typical peer was instructed to provide a verbal prompt (e.g., "Kenny, it's MY turn now"). The educator reminded the peers to use these strategies during each intervention session.

Procedures

Ethics

Informed written consent for the children to participate in the research was provided by their parents. Peers were encouraged to participate in individual sessions but were able to refuse if they did not wish to join a particular session. The children with disabilities were encouraged to participate in the sessions but were allowed to leave if they wished. As required by the university ethics committee, all video data were stored on password-protected computers available only to the researchers.

Training

Scripts for educator implementation of (a) procedures for the conduct of the baseline, intervention, and maintenance sessions and (b) peer training and coaching were developed by the research team and modified following trials using children not included in the research.

Educator training

Two educators participating in the research practiced using the training and coaching scripts with the third author over two 1-hr sessions. Children who were not part of the

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research were included in the practice sessions. These sessions were filmed, with footage used to provide feedback to the educators.

Peer training and coaching

Peers were trained by their educator to support the child with ASD in taking turns playing a game on an iPad and coached by the educator during the intervention sessions to provide appropriate prompts to their turn-taking partner. Training and coaching were delivered using scripts implemented by the educator working in the room in which the children with and without disabilities were enrolled. Two scripts were used for training and one for coaching. For the initial training of each typical peer prior to the first intervention session in which he or she participated, the script provided detailed modeling of the intervention and opportunities for peers to practice using prompts. During subsequent training sessions, a second script was used to remind peers (a) how they should encourage their friend to take turns with the app and (b) that the educator was available to remind them what to do.

Training of peers was conducted by educators in the indoor play area just prior to each intervention session, which was implemented in the same space. At this time, the other children were involved in scheduled outdoor play. Initial training sessions (11 sessions) lasted a mean of 2 min 13 s (range, 1 min 37 s–3 min 12 s). Subsequent training sessions (33 sessions) lasted a mean of 1 min 35 s (range, 1 min 11 s–2 min 38 s). Because most sessions lasted for up to 8 min (peer training plus intervention), the children were still able to participate in outdoor play. Peers were chosen to support the target child based on the availability on the day of a child for whom parent permission had been provided.

Collection of turn-taking data

Data collection began in late September and finished in late December just prior to the Christmas break. All sessions (baseline, intervention, and maintenance) were implemented in the indoor play area while the other children participated in outdoor play between 10 a.m. and 10.30 a.m. This allowed filming of the sessions to occur without interruption and for the participants to join outdoor play when the session was completed. All sessions were of 5-min duration.

Coding turn-taking behavior

An event-recording system was developed by the research team to record the turn-taking behavior of the child with ASD using the video footage for each session. The first author and a research assistant simultaneously coded data from a range of sessions until they achieved reliability of at least 80% (after ~2 hr). The research assistant then coded the turn-taking behavior for all sessions included in the research. Over the 5-min period, data were coded for the first 20 turns, with the possibility of 10 successful turns being recorded for each child (target and peer). Fewer than 20 turns were recorded for only three sessions: the third baseline session for Adam (18 turns) and the second and fourth intervention sessions for Noah (12 and 14 turns, respectively).

Study implementation

The three session types included in the study were baseline, intervention, and maintenance. There were two intervention phases for Adam.

Baseline

The educators acting as facilitators asked the target child (i.e., the child with ASD) and a peer to play a game using the iPad. The children were seated side by side, with the iPad on a mat between them. The Peekaboo app was open on the iPad ready to use. The iPad was locked so that neither child could close the iPad game. The children were told that the iPad must remain on the mat where both could reach it. The only intervention by the educators was to replace the iPad if it were removed and to remind the children that the iPad must remain on the mat.

Five baseline sessions were implemented for the first child (Kenny) before the

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intervention was introduced. At the same time, three baseline sessions were implemented for the second child (Adam) and the third child (Noah). Two additional baseline sessions were implemented for Adam and Noah just prior to the introduction of their interventions. The intervention was introduced to Adam after an intervention effect was observed for Kenny (after Session 6) and to Noah after an intervention effect was observed for Adam (after Session 11).

Intervention

Intervention sessions were implemented each time the target child attended the center (i.e., one to three times a week depending on the child's attendance). Each turn-taking session was preceded by a short training session, in which the educator modeled (during the peer's first play session with the target child) and rehearsed (in subsequent training sessions) the prompts that the peer needed to use to assist the target child to take his turn.

For each intervention session, the educator sat behind the children who were seated side by side, with the iPad located on a table in front of them. He or she was to remind the children that the iPad was to stay on the mat, which was located between the two children. For each opportunity for the target child to take a turn, educators were to wait for the 3-s latency period to finish. If the peer had not begun to provide the appropriate prompt after another 2 s, the educator was to ask the peer to (a) tell the target child that it is his or her turn or (b) help the target child take his or her turn, whichever was appropriate. If the peer began to prompt the child without waiting, the educator was to remind him or her that he or she needed to wait. If either child did not allow the other to take his or her turn, the educators were to provide a verbal prompt and if necessary a physical prompt (i.e., blocking the turn) to ensure that the correct turn was taken.

A second intervention phase was included for Adam because he was having difficulty sharing turns with his peer. A modification was made to the training script, with the educator instructing the typical peers to take their turns quickly when it was their turn without providing the verbal reminder that it was their turn. This was to be followed up, when necessary, through the coaching, which was implemented during the intervention sessions.

Maintenance

Maintenance probes were implemented once the target children first demonstrated a success rate of 80% for turn taking. These probes were conducted once a week for Kenny, and only once for Adam, on the same day but prior to the intervention session. The same or a different peer to the one supporting the intervention on that day was included in the maintenance session. The conditions for the maintenance sessions were the same as those implemented for the baseline sessions (i.e., no peer training or coaching provided by the educators). For these sessions, the educator sat behind the children and ensured that the iPad remained on the mat between the children.

Treatment fidelity

Data sheets used to measure procedural fidelity for the implementation of the training and coaching procedures by the educators and the implementation of the intervention by the peers were developed by the researchers based on the scripts used to train and coach the peers. Following a joint trial of the measures with the first author, the research assistant measured the same intervention sessions for educator procedural fidelity and the same intervention and maintenance sessions for peer procedural fidelity as were used to calculate interrater reliability for turn taking. These sessions, along with 20% of baseline sessions used in the measure of interrater reliability for turn taking, were selected randomly using the website www.RANDOM.ORG. A research assistant, who was the primary observer, coded at least 20% of the sessions for procedural fidelity.

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Educator procedural fidelity

Educators were scored for their delivery of the scripted training (*always*, *mostly*, *sometimes*, *rarely*, *never*), implementation of the session opening and closing procedures (yes/no), and use of scripted coaching procedures during the interventions sessions. The coaching procedures that were measured included (a) waiting for the peer to deliver the appropriate prompt; (b) prompting the peer to provide an appropriate prompt ("your turn"/physically assisting the turn); (c) reminding the peer to wait; (d) prompting the peer to take his or her turn; (e) blocking an incorrect turn; and (f) reminding the peer to take his or her turn quickly (second intervention for Adam). It was important for the educator to provide coaching procedures only as required to avoid unnecessary prompting. Therefore, for all coaching procedures, except for waiting to prompt, fidelity was calculated for each turn-taking event. The number of turn-taking events for which appropriate coaching strategies and appropriate noncoaching strategies were observed was divided by the number of appropriate coaching/noncoaching incidents and inappropriate coaching/noncoaching incidents and multiplied by 100.

Fidelity for waiting to prompt was calculated by recording appropriate waiting if the educator reminded the peer to prompt the target child's turn after the prescribed wait time (3 s + 2 s). The percentage of fidelity for this coaching strategy was calculated by dividing the number of positive instances of waiting by the number of positive and negative instances and multiplying by 100.

Peer procedural fidelity

The research assistant recorded a positive or negative score for each opportunity that the peer had to (a) take his or her turn or (b) allow the target child to take his turn. The percentage of peer fidelity for taking his or her turn or allowing the target child to take his turn was calculated by dividing the number of times that the peer took a turn or allowed the target child to take a turn by the number of presenting opportunities and multiplying by 100.

It was important for the peer to provide prompts only as required in order to avoid unnecessary prompting. Therefore, for all prompts, fidelity was calculated for each of the turn-taking events. Prompts considered unnecessary or prompts provided without waiting for the 3-s latency period were scored as inappropriate. The number of times that appropriate prompts and nonprompts was observed was divided by the number of incidents of appropriate plus inappropriate prompts/nonprompts and multiplied by 100.

Fidelity for waiting to prompt was calculated by scoring appropriate waiting if the peer prompted the target child to take his turn after the prescribed wait time (i.e., 3 s). The percentage of fidelity for waiting to prompt was calculated by dividing the number of positive instances of waiting by the number of positive and negative instances and multiplying by 100.

Interrater reliability

Measures of interrater reliability were collected for (a) successful turn taking, (b) measures of the fidelity with which educators implemented the training and coaching procedures, and (c) peers sharing turns and use of prompts to assist with turn taking.

Turn taking

In addition to the practice undertaken before turn-taking behavior was coded using the video footage, at least 20% of the sessions were independently coded by the first author to measure interrater reliability for successful turn taking. One session from the baseline for each of the three target children was included for a reliability check and at least 20% of intervention sessions for each child were also included: four of the 20 sessions for Kenny, four of the 19 intervention sessions for Adam (two of the nine Phase 1 intervention sessions and two of the 10 Phase 2 intervention sessions), and one of the five intervention sessions for Noah. For Kenny, two of the seven

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maintenance sessions were included for reliability checks and, for Adam, the one maintenance session was also included.

Interrater reliability was calculated by dividing the number of agreements by the number of agreements plus disagreements and multiplying by 100. Mean interrater reliability was 95.7% (range, 85%–100%). In almost every case of disagreement, the target child rested his fingers on the bottom edge of the iPad, making it very difficult to determine whether or not he was trying to take his peer's turn. See Table 2 for the percentage of interrater reliability for each session included for a reliability check.

Educator procedural fidelity

The nine sessions coded by the research assistant for procedural fidelity were also coded by the first author to allow for a measure of interrater reliability. For level of script use (*always*, *mostly*, *sometimes*, *rarely*, *never*), implementation of opening and closing procedures (yes/no), and coaching procedures, reliability was calculated by dividing the agreements by the agreements plus disagreements and multiplying by 100.

Both observers agreed that appropriate opening and closing procedures were implemented by the educators for each session. For six of the nine sessions, both observers provided identical ratings for the use of the script (66.7%). There was disagreement for three of the sessions, but for each of these, the observers differed by only one rating level (see Table 3). Both raters agreed that the number of opportunities to use coaching strategies was low for waiting and prompting to wait. Urging the peer to take his or her turn quickly was measured for two of the nine sessions, as this coaching strategy was implemented only for Adam during the second intervention phase. The mean reliability across all coaching strategies was 91%. As can be seen in Table 3, interrater reliability was low in sessions I11 (Kenny) and I2 (Adam) for both prompting the peer to take his or her turn and verbally or physically blocking an inappropriate turn. Lower reliability for these sessions was possibly due to the difficulty hearing what the educators were saying.

Peer procedural fidelity

The first author coded the same sessions for peer procedural fidelity as coded by the research assistant. For each turn-taking event, a measure of interrater reliability was then calculated for peers (a) taking their own turns and allowing the target child to take a turn; (b) providing verbal and physical prompts to the target child; and (c) waiting before verbal or physical prompts were provided. Reliability was calculated by dividing instances of the agreed presence/absence of prompting/waiting by agreements plus disagreements and multiplying by 100. Interrater reliability for peer fidelity was 94.6% for taking a turn and 94% for allowing the target child to take a turn (see Table 4 for interrater reliability of peer fidelity across target children).

Table 2. Percentage Interrater Reliability for Turn Taking

	Baseline		Intervention 1		Intervention 2		Maintenance	
Kenny	B5	95	I9	100			Ma4	95
			I11	100			Ma5	100
			I13	100				
			I17	95				
Adam	B3	100	I2	100	I10	90	Ma1	85
			I7	95	I11	95		
Noah	B4	100	I5	85				

Note. B = baseline; I = intervention; Ma = maintenance.

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Table 3. Percentage Interrater Reliability for Educator Procedural Fidelity

		Use of Script			Agreed Use of Coaching Strategies				
Target Child	**Session**	**Prim**	**Sec**	**Wait**	**Prompt Peer to Prompt Turn**	**Prompt Peer to Wait**	**Prompt Peer to Take Turn**	**Block Incorrect Turn**	**Prompt Peer to Take Turn Quickly**
Kenny	I9	S	S	100 (0)	95 (1)	100 (0)	95 (5)	100 (2)	N/A
	I11	S	S	100 (0)	100 (0)	100 (0)	60 (11)	60 (11)	N/A
	I13	M	M	100 (0)	100 (0)	100 (0)	100 (3)	90 (2)	N/A
	I17	S	M	100 (0)	100 (1)	100 (0)	90 (6)	90 (6)	N/A
Adam	I2	M	A	100 (0)	100 (0)	100 (0)	50 (17)	60 (9)	N/A
	I7	M	M	100 (0)	90 (2)	95 (1)	90 (3)	95 (9)	N/A
	I10	M	M	100 (0)	100 (0)	100 (0)	95 (1)	80 (10)	85 (11)
	I11	M	M	100 (0)	100 (0)	95 (1)	95 (3)	90 (3)	80 (10)
Noah	I5	M	S	90 (2)	90 (2)	95 (7)	95 (1)	85 (3)	N/A
Mean % agreement		66.7		98.9	97.2	98.3	85.6	83.3	82.5

Note. A = always; I = intervention; M = mostly; N/A = not applicable; Prim = primary observer; S = sometimes; Sec = secondary observer. Percentages calculated correct to the first decimal point. The number of prompt opportunities for which there was both agreement and disagreement is given in parentheses.

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Table 4. Percentage Interrater Reliability for Peer Procedural Fidelity

		Takes Own Turn		Allows Target Turn	
Target Child	Number of Sessions	Total Opportunities (Range Across Sessions)	Mean % Agreement (Range Across Sessions)	Total Opportunities (Range Across Sessions)	Mean % Agreement (Range Across Sessions)
Kenny	6	61 (9–11)	98 (90–100)	61 (9–11)	97 (89–100)
Adam	5	57 (10–14)	90 (79–100)	46 (8–10)	91 (75–100)
Noah	1	11	100	9	100
		Verbal Prompt "My turn"	Verbal Prompt "Your Turn"	Physical Prompt	Wait to Prompt
		Mean % Agreement (Range Across Sessions)	Mean % Agreement (Range Across Sessions)	Mean % Agreement (Range Across Sessions)	Mean % Agreement (Range Across Sessions)
Kenny	6	89 (78–98)	93 (80–100)	100	100
Adam	5	91 (85–98)	96 (93–98)	99.5 (98–100)	99.5 (98–100)
Noah	1	98	95	100	99

Note. Percentages calculated to the nearest whole. The number of opportunities includes all opportunities for which there was agreement and disagreement. Wait to prompt includes waiting before physical and verbal prompts.

Only one rater, the second observer, recorded a physical prompt in the sample of sessions used for interrater reliability. Although both observers noted many opportunities for the peer to provide verbal prompts to the target child, verbal prompts were observed by both raters in only three of the 12 sessions in which interrater reliability was measured (Adam: I2, I7; Noah: I5).

Interrater reliability was 90.4% for the presence/absence of verbal prompts to the target child to allow the peer to take a turn and 94% to allow the target child to take his own turn. Interrater reliability was 99.7% for appropriate waiting before providing a verbal or physical prompt to the target child to take his turn.

RESULTS

Turn taking

As illustrated in Figure 1, the baseline data for each of the three participants were relatively stable, with turn taking present for 35% or fewer opportunities for all baseline sessions (range, 5%–35%). A functional relationship between the introduction of the intervention and increased turn taking for Kenny and Noah was demonstrated immediately or soon after the introduction of the intervention, with a continuing upward trend. The percentage of successful turn-taking behavior during the intervention phase for Kenny fell below baseline for only two of the 19 sessions and reached 100% toward the end of the intervention. He was able to achieve an above baseline performance for six of the seven maintenance sessions. The percentage of nonoverlapping data for Kenny's turn-taking sessions following the introduction of the intervention was 88.5%. The percentage of successful turn-taking behavior was well above baseline measures for all of Noah's intervention sessions.

Although there was some improvement in Adam's turn-taking behavior following the

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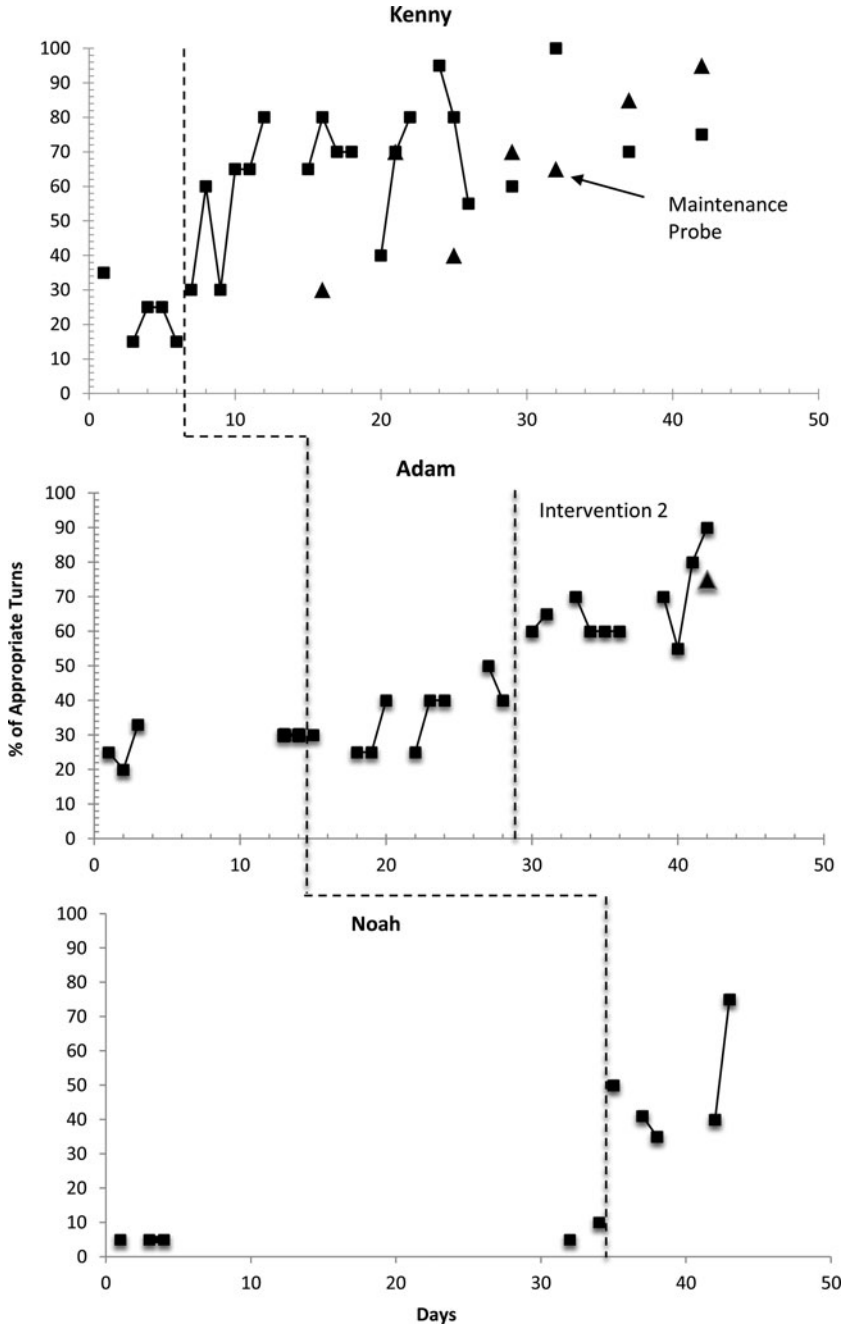


Figure 1. Successful turn taking.

introduction of the first intervention, only five of the eight data points were above baseline level (62.5%) and there was only a small upward trend. With the introduction of the second intervention phase, there was a steeper upward trend and all data points were above baseline level. Maintenance data were collected only once for Adam, with successful

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turn taking at 75% for maintenance being well above the top baseline score of 33%.

Educator procedural fidelity

Training and coaching scripts were not used in baseline sessions or for maintenance probes. They were, however, used consistently during the intervention phases but not always read word for word. The educator for two of the children, Sue, was rated by the research assistant as mostly following the script for two of the sessions and sometimes following the script for three of the sessions coded for educator procedural fidelity. Improvisation sometimes resulted in parts of the script being omitted altogether. The educator for the third child was rated as mostly following the script for all four sessions coded.

The setup and closing procedures were consistently implemented. The setup involved seating the children side by side, placing the iPad on the mat between the children, and reminding the children that the iPad must remain on the mat. Closing involved stating that all the animals had been seen and thanking the children for their participation. Although the script for the closing comments was not strictly followed, both educators provided a closing comment in each of the sessions included for fidelity checking.

Eight different peers were included in the sessions observed for educator procedural fidelity. Implementation of coaching strategies varied across the types of coaching strategy (see Table 5). Fidelity for appropriate waiting before prompting was 66.7%. For the other coaching strategies, the mean percentage for fidelity of implementation was 95.7%.

Peer procedural fidelity

In Table 6, peer procedural fidelity data are provided for the individual sessions coded. Peers successfully took their turns 71.4% of the time and allowed the children with ASD to take their turns 85.8% of the time. These data suggest that peers were generally more successful in allowing the child with ASD to take a turn than they were in taking their own turns. During the first intervention with Adam, his peers had difficulty taking turns, as Adam took almost every turn. With the additional coaching component provided to peers by the educator in the second intervention, turn taking was much more evenly shared across the two children.

Prompts by peers were recorded as appropriate when provided spontaneously or when provided following coaching by the educators. Peers generally prompted only when it was appropriate to do so (use/nonuse of verbal prompts, 90% of the turn-taking events coded; use/nonuse of physical prompts, 100% of the turn-taking events coded). Peers, therefore, demonstrated appropriate use/nonuse of verbal and physical prompts 95% of the time. However, waiting before providing the target child with a verbal prompt to take his turn was less commonly observed (55.6% of the presenting opportunities). No physical prompts were needed or provided in the sessions coded for peer procedural fidelity.

DISCUSSION

Treatment effect for Peer-mediated intervention

Visual inspection of the data indicated an intervention effect for all three children with ASD. Furthermore, both Kenny and Adam, for whom maintenance data were collected, were able to maintain high levels of turn taking during the maintenance sessions. This suggests that the level of turn-taking skills demonstrated in the intervention sessions could be maintained, even though training and coaching of peers by the educators were withdrawn.

Noah's improvement in turn taking from the baseline to intervention phase was the most striking of all three children. He was the last child to be included in the study and, judging by his low baseline performance and the time left before the Christmas break, expectations for a treatment effect were quite low. During the baseline phase, Noah remained standing and cried for much of the time. He would pull at the hand of the educator to

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Table 5. Percentage Educator Procedural Fidelity for Coaching Strategies

Target Child	Session	Waits to Prompt	Prompt Peer to Prompt Turn	Prompt to Wait	Prompt Peer to Take Turn	Block Incorrect Turn	Prompt Peer to Take Turn Quickly
Kenny	I9	100 (1)	100 (1)	100 (0)	80 (4)	90 (2)	N/A
	I11	[a]	100 (0)	100 (0)	90 (5)	80 (11)	N/A
	I13	[a]	100 (0)	100 (0)	100 (3)	100 (0)	N/A
	I17	0 (1)	95 (1)	100 (0)	75 (5)	70 (6)	N/A
Adam	I2	[a]	100 (0)	100 (0)	75 (11)	90 (4)	N/A
	I7	0 (1)	95 (2)	95 (1)	100 (1)	100 (9)	N/A
	I10	[a]	100 (0)	100 (0)	100 (0)	100 (2)	100 (10)
	I11	[a]	100 (0)	100 (1)	95 (3)	100 (0)	100 (9)
Noah	I5	100 (2)	100 (2)	80 (6)	100 (1)	95 (1)	NA
Total opportunities		5	6	8	33	35	19
Mean % reliability		66.7	98.9	97.2	90.6	91.7	100

Note. I = intervention; N/A = Not applicable. Percentages calculated correct to the first decimal point. The number of times that a prompt was needed or waiting was required is given in parentheses.

[a]No opportunities were observed.

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Table 6. Peer Procedural Fidelity

				Verbal Prompting "My Turn" "Your Turn"			Physical Prompt		
Target Child	**Session**	**Takes Own Turn, % Successful**	**Allows Target Turn, % Successful**	**Number of Prompts Given**	**% Appropriate Use/Nonuse**	**% Use of Wait Time[a]**	**Number of Prompts Given**	**Appropriate Use/Nonuse**	**% Use of Wait Time[a]**
Kenny	I9	90 (10)	90 (10)	0	98.8	[b]	0	100	[b]
	I11	36.4 (11)	62.5 (8)	0	87.5	[b]	0	100	[b]
	I13	72.7 (11)	100 (9)	0	92.5	[b]	0	100	[b]
	I17	60 (10)	80 (10)	0	91.3	[b]	0	100	[b]
	Ma 4	90 (10)	70 (10)	0	91.3	[b]	0	100	[b]
	Ma5	66.7 (9)	63.6 (11)	0	87.5	[b]	0	100	[b]
Adam	I2	16.7 (12)	75 (8)	3	76.3	[b]	0	100	[b]
	I7	81.8 (11)	100 (9)	10	95	100 (2)	0	100	[b]
	I10	100 (10)	100 (10)	0	91.3	[b]	0	100	[b]
	I11	90 (10)	90 (10)	0	86.3	[b]	0	100	[b]
	Ma1	72.7 (11)	100 (9)	0	93.3	[b]	0	100	[b]
Noah	I5	90 (11)	100 (9)	9	90	22.2 (9)	0	100	[b]
Total opportunities		126	113	22		36	0	100	
Mean % reliability		71.4	85.8					100	

Note. I = intervention; Ma = maintenance. Percentages correct to the first decimal point. The number of opportunities is given in parenthesis.

[a]Relative to opportunity.

[b]No verbal/physical prompts provided.

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try to get her to move away from the table and would frequently move away from the turn-taking game only to come back again to secure the educator's attention. This behavior changed dramatically once his peers were taught to support his turn-taking behavior. He sat for the duration of four of the intervention sessions, leaving one session to take a toilet break. Noah was in the room while the training of the peer took place but was not asked to sit with the peer. At times, he would come to the table while the peer was being instructed by the educator and appeared to show some interest in the process. This may explain why he sat side by side with his peer from the first intervention session.

Adam's performance following the introduction of the initial intervention did not demonstrate a clear treatment effect. Because he commonly took another turn immediately after his turn, even when his peer called out "My turn Adam," the educator began to block Adam from taking a turn inappropriately. This was scored as unsuccessful turn taking as Adam was trying to take the peer's turn. As this behavior did not change over several sessions, a new feature was introduced into the intervention, constituting a second intervention phase for Adam. In the training sessions, Adam's peers were now told to take their turns quickly. This direction to peers was also introduced into the coaching component of the intervention. Following the introduction of the second intervention, Adam's turn-taking behavior increased to percentages above those demonstrated in the baseline and first intervention phases and continued with an upward trend.

Educator procedural fidelity

Cook and Odom (2013) identified the difficulty integrating evidence-based practices into the everyday teaching of educators. Although fidelity is identified as a critical element to support the widespread use of evidence-based practices by educators, it is unclear how much adherence to all aspects of an intervention is required in order to achieve an outcome (Strain, 2018). In the current study, educators used appropriate waiting strategies for only 66.7% of the opportunities recorded. Although this is a low percentage, a robust effect was still observed in all three children. This suggests that adherence to the exact wording of the script may not be the salient aspect of this intervention. Although this appeared to be the most challenging strategy required of the educators, the low score for this coaching component would also have been impacted by the relatively low number of presenting opportunities. Mean use of the remaining coaching strategies by the educators was high at 95.7%. In this study, educators were able to implement an intervention, which had been identified in the research literature as a promising practice, with sufficient reliability to ensure that there was an opportunity for the intervention to effect change.

Peer procedural fidelity

Although peers were more likely to allow the children with ASD to take their turns than they were to take their own turns, there was an increase in successful turn taking for all children with ASD following the introduction of the coaching intervention. Peers prompted appropriately for 95% of the time during the sessions for which procedural fidelity was measured, but they did not always wait before providing a prompt. Despite the fact that peers did not always follow prompting procedures as instructed, their support for the children with disabilities appears to have been sufficient to promote turn-taking behavior.

It is possible that some peers will be more successful than others in supporting an intervention. The selection of peers to support the turn-taking skills of the children with ASD was based on the return of permission notes by parents and the presence of the peers on the days that the children with disabilities attended the centers. Two peers supported Noah, whereas four and five peers supported Adam and Kenny, respectively. Perhaps, greater opportunities for practice by having only one peer supporting each target child might have resulted in higher levels of peer fidelity, in particular waiting before providing

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a prompt. However, this would not have been practical, given that different children attended on different days. Also, the absence of that one peer may have meant that the intervention could not be implemented on some days.

Implications for practice

The value of using peers to support children with disabilities in inclusive childcare settings is evidenced by all three children with ASD improving and maintaining their turn taking despite imperfect fidelity of implementation by both educators and peers. If interventions are to have an effect in the real world, which is, after all, the aim of researchers hoping to improve instruction for children with disabilities and other special needs, research is required that demonstrates that interventions can work when less rigor is applied.

Research into peer-mediated practice has generally used research assistants or highly trained educators to train and facilitate typical peers to implement interventions (Katz & Girolametto, 2013). In the current research, the children's own educators implemented the intervention with minimal training (2 hr) by the researchers. The researchers advised on a promising practice for children with disabilities, and the practitioners cooperated in implementing the research in a naturalistic setting, thereby allowing the researchers to test the effectiveness of the intervention in an applied setting. The collaboration between researchers and practitioners had advantages for both professions (Strain, 2018).

Every effort was made to strengthen the internal validity of the research by using a strong single-case design and maximizing the reliability of the coding and the fidelity of the implementation of the intervention by training the educators and peers using carefully developed training and coaching scripts. Although it is acknowledged that the use of strategies that have a strong evidence base will not necessarily work for every child (Cook & Odom, 2013), research evidence can and should be used to guide practice in early childhood intervention. In this case, the intervention was implemented in an inclusive setting, where such practice is not commonly observed. Despite regular changes in peer partners, the gains in turn-taking skills made by the children further supported the value of this type of intervention in an inclusive setting. It is possible that the use of several peers assisted with generalization.

Limitations of the research

Research can be difficult to implement in natural environments. Using educators to implement interventions integral to the research can provide a threat to internal validity. There is a good chance that the interventions will not be implemented as planned if educators forget to follow the script, are distracted, or fail to see the importance of components of the intervention. In the current research, not all components of the intervention were well implemented by the educators. What is not known is whether a more reliable implementation on the part of the educators would have resulted in even higher levels of turn taking by the children with ASD.

Given the applied nature of the research and the implementation of the intervention by the children's educators, it would have been valuable to include a measure of social validity as a component of the research. If the intention is to encourage practitioners to use evidence-based interventions in their daily practice, it is important to know that they not only see value in the intervention but are confident in their ability to implement it as well. Future research of this nature should include a measure of social validity in addition to the measure of intervention effect.

The current study demonstrated the effectiveness of using regular peers to support the turn-taking skills of three children with a diagnosis of ASD and other disabilities. Turn taking is an important social skill and one that was valued by both families and educators. Joint attention would ordinarily be a prerequisite skill for turn taking. However, in the current study, joint attention with the peer while taking his turn with the iPad was not necessary for the target child to be scored

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as successfully taking his turn or allowing his peer to take a turn. Given the importance of joint attention to social interaction and communication, it would be valuable to establish whether (a) there was a relationship between joint attention and turn taking, (b) joint attention preceded gains in turn taking, or (c) developing skills in turn taking could promote joint attention where this was not present. Because turn taking was a priority skill for the families and this was one of the priority objectives for each child's individual education plan, the relationship between joint attention and turn taking was not explored in the current study.

As with all single-case research, external validity is limited. It is important, therefore, that this research is replicated with other participants, those with ASD, and those with other disabilities or special needs. In addition, replication of this research will be important in other settings and contexts, such as those in other countries in which personnel training and inclusive early childhood programs may not be as available.

CONCLUSION

In the current study, three children with ASD demonstrated improved skills in turn taking following a peer-mediated intervention facilitated by educators in two inclusive childcare centers. Although neither the peers nor the educators implemented the intervention with perfect fidelity, the intervention was robust enough to support the development of these important social interaction skills. The success of the intervention demonstrates the importance of a cooperative relationship between researchers and practitioners if the gap between research and practice is to be addressed.

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